

Explicit Andrews–Curtis constructions for divisibility subfamilies of square presentations

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Abstract

We give finite, explicitly witnessed ordinary Andrews–Curtis constructions for the ordered presentations

$$G(p, s) = (u^2 y^{-(2s+1)}, uy^p uy^{-s} u^{-1} y^{-p}).$$

For natural numbers p, s , the construction applies whenever $p \mid s$ or $p \mid (s+1)$; in particular, it covers $G(2, s)$ at every nonnegative height. A fixed route enters a root-conjugacy presentation, after which an iterated doubling witness supplies either a power elimination or a reduction of the root degree. Every substitution carries an ordered product of conjugates of the other relator and restores that relator. The quantified theorems are formalized in Lean using the SAIR ordinary AC relation. A separate exporter produces numbered certificates for independent replay. We also state an exact residual reduction and the unresolved companion-restoration condition for $G(3, 3m+1)$. The remaining square family and the general conjectures are not settled by these results.

1 Statement and context

Work in the free group $F = F(u, y)$, with ordered pairs of relators. Write $R \sim_{\text{AC}} S$ when one can pass from R to S by finitely many relator inversions, right multiplications by the other relator, and conjugations. The rank is always two. Put $\text{Sol}(R)$ for $R \sim_{\text{AC}} (u, y)$. Conjugation by an arbitrary word is counted as one mathematical primitive; the executable certificates expand it into conjugations by letters. Multiplication by the inverse of the other relator costs three primitives, including the two inversions that restore it.

Theorem 1.1 (Divisibility construction). *Let $p, s \in \mathbb{N}$, where $0 \in \mathbb{N}$. If $p \mid s$ or $p \mid (s+1)$, then $\text{Sol}(G(p, s))$.*

The proof constructs the moves without assuming that the presented group is trivial. Thus triviality is also a consequence. The admissible case $p = 0$ in this theorem is precisely $s = 0$.

Corollary 1.2. *For every $s \in \mathbb{N}$, $\text{Sol}(G(2, s))$.*

Proof. One of $s, s+1$ is divisible by two. □

Our point of comparison is the Miller–Schupp family

$$\text{MS}(n, w) = \langle x, y \mid x^{-1} y^n x = y^{n+1}, x = w \rangle,$$

where $n > 0$ and w has exponent sum zero on x . Shehper et al. prove AC-triviality for all $\text{MS}(1, w)$, for all $\text{MS}(n, y^{-1}xyx^{-1})$, and for $\text{MS}(2, y^{-k}x^{-1}yxy)$ for every integer k [1, Theorem B; Theorems 2,

3 and Corollary 9]. Their substitution operation is an explicit predecessor of the witness operations used here. The earlier $p = 1$ analysis in this project enters their $\text{MS}(1, w)$ class. The proof below supplies its own root-family witnesses rather than importing that theorem as an assumption.

Substitution by conjugate products and transport under automorphisms are established techniques; see Panteleev–Ushakov [2]. Lackenby’s ordinary AC theorem for thickenable balanced presentations gives a different sufficient condition [3, Theorem 1.3]. We have not determined how the entire family here intersects that geometric class. Lean formalization of AC and automated certificate checking also have prior implementations [4, 6]. The contribution presented here is this explicit arithmetic construction and its formal proof package. We make no priority claim for the underlying AC-triviality classes or the general verification methods.

2 Witnesses that compile to ordinary moves

Definition 2.1. A witness relative to a relator r , from a to b , is a finite ordered list $W = ((g_1, \epsilon_1), \dots, (g_t, \epsilon_t))$, with $\epsilon_i \in \{1, -1\}$, together with the identity in the ambient free group

$$a \prod_{i=1}^t g_i r^{\epsilon_i} g_i^{-1} = b. \quad (1)$$

We write $a \xrightarrow[r]{W} b$.

This contains more data than an equality in the presented quotient.

Lemma 2.2 (Witness compilation). *An identity (1) gives $(a, r) \sim_{\text{AC}} (b, r)$, with the second relator restored after each factor. A positive factor costs three mathematical primitives and a negative factor costs five. The analogous assertion holds with the two slots exchanged.*

Proof. For a positive factor, conjugate the donor r by g_i , right multiply the recipient, and conjugate the donor back by g_i^{-1} . For a negative factor, invert the donor before the multiplication and invert it again afterward. Repeat in the listed order. $\square$

For completeness, the witness operations used below are explicit. Composition concatenates factor lists. Conjugation by h replaces every g_i by hg_i . Given $a \Rightarrow b$ and $c \Rightarrow d$ relative to the same donor, a witness $ac \Rightarrow bd$ is obtained by replacing each conjugator in the first list by $c^{-1}g_i$, then appending the second list. To obtain $a^{-1} \Rightarrow b^{-1}$, reverse the list, negate all signs, and replace each conjugator by ag_i . Iterating the product operation constructs positive powers; inverse words give negative powers. These rules follow directly by multiplying (1) and do not require a witness extraction oracle.

3 Entry into a root-conjugacy pair

Set

$$B_p = y^{-p} u^2 y^p u^{-1}, \quad A_s = y^{-(s+1)} u^{-1} y^s u^{-1}. \quad (2)$$

Lemma 3.1 (Uniform entry). *For all integers p, s , a route of 61 mathematical primitives takes $G(p, s)$ to $(f(B_p), f(A_s))$, where*

$$f(u) = y^{s+1-p} u^{-1} y^p, \quad f(y) = y^{-1}.$$

We have $\text{Sol}(G(p, s))$ if and only if $\text{Sol}((B_p, A_s))$.

Proof. The complete route is in Appendix A. After its first twelve macros, which cost eighteen primitives, the pair is

$$(y^{-p}u^{-1}y^p u^{-1}y^s u, \quad uy^{-p}u^{-2}y^p uy^{-1}).$$

The remaining macros cost 43 primitives and end at the displayed image pair. Each step is a free-group identity, independently checked in the accompanying Lean files `ACDiagonalDescent` and `ACSquareP2Entry`.

The map f is an automorphism: its inverse sends u to $y^{-p}u^{-1}y^{p-s-1}$ and y to y^{-1} . A homomorphism sends every AC step to an AC step, with conjugators mapped by that homomorphism. To finish a transported solution at $(f(u), f(y))$, conjugate its first relator by y^{p-s-1} , obtaining $u^{-1}y^{s+1}$; right multiply it by the second relator to power $s+1$, then invert both relators. For the inverse image basis, conjugation by y^p similarly gives $u^{-1}y^{-s-1}$; right multiplication by the second relator to power $-s-1$, then two inversions, finishes. These explicit basis corrections and reversibility of the displayed route give both implications. $\square$

The change of coordinates in this argument transports a complete path and includes its final correction; it is not inserted as an additional AC move.

4 The two arithmetic finishes

Lemma 4.1 (Iterated doubling). *For $p \in \mathbb{Z}$ and $m \in \mathbb{N}$, there is a supplied witness relative to B_p*

$$y^{pm}uy^{-pm} \implies u^{2^m}.$$

The recursive list described here has $2^m - 1$ factors.

Proof. The basic witness from $y^p uy^{-p}$ to u^2 has the single positive factor with conjugator $y^p u^{-1}$; explicitly,

$$(y^p uy^{-p})(y^p u^{-1})B_p(y^p u^{-1})^{-1} = u^2.$$

Start at $m = 0$ with the empty witness. At stage $m + 1$, conjugate the stage- m witness by y^p , then compose with the 2^m -th power of the basic witness. The resulting target is $u^{2^{m+1}}$. The factor count satisfies $L_0 = 0$ and $L_{m+1} = L_m + 2^m$, proving the formula. $\square$

Lemma 4.2 (Power finish). *For any integers k, e , $\text{Sol}((B_k, y^{-1}u^e))$.*

Proof. With donor $q = y^{-1}u^e$, the identity $yq = u^e$ supplies a one-factor witness replacing y by u^e . Apply its power and product witnesses to $B_k = y^{-k}u^2y^k u^{-1}$ to obtain u . The donor is restored. Right multiply the second relator by the first to power $-e$ to obtain y^{-1} , then invert. $\square$

Theorem 4.3 (Arithmetic root families). *For all $p \in \mathbb{Z}$ and $m \in \mathbb{N}$, both*

$$\text{Sol}((B_p, A_{pm})) \quad \text{and} \quad \text{Sol}((B_p, A_{pm+p-1}))$$

hold by finite witness constructions.

Proof. Put $d = 2^m$. In the first case, conjugate the companion by y^{pm} :

$$y^{pm}A_{pm}y^{-pm} = y^{-1}u^{-1}(y^{pm}u^{-1}y^{-pm}).$$

Use the inverse-word witness of Lemma 4.1 in the parenthesized factor. This leaves $y^{-1}u^{-(d+1)}$, to which Lemma 4.2 applies.

In the second case, the same conjugation and witness leave

$$q = y^{-p}u^{-1}y^{p-1}u^{-d}. \quad (3)$$

Writing $v = u^{-1}y^{p-1}u^{-d}$, the exact identity $y^p q = v$ is a one-factor witness relative to q . Its inverse-word and product witnesses replace the two occurrences of y^p and its inverse in B_p , giving

$$v^{-1}u^2vu^{-1} = u^dB_{p-1}u^{-d}.$$

Conjugate this first relator by u^{-d} . The pair is now (B_{p-1}, q) . Conjugate q by y^{p-1} and rewrite it as

$$y^{p-1}qy^{-(p-1)} = y^{-1}u^{-1}(y^{p-1}u^{-d}y^{-(p-1)}).$$

The $(-d)$ -th power of the basic degree- $(p-1)$ doubling witness changes this to $y^{-1}u^{-(2d+1)}$. Apply Lemma 4.2. At each substitution, Lemma 2.2 restores the donor, so the concatenation is a path of ordinary rank-two moves. $\square$

Proof of Theorem 1.1. If $s = pm$ with $m \in \mathbb{N}$, use the first case of Theorem 4.3 and Lemma 3.1. If $s + 1 = pk$ with $p, s, k \in \mathbb{N}$, then $k \geq 1$; set $m = k - 1$, so $s = pm + p - 1$, and use the second case. This includes the degenerate $p = s = 0$ through the first case. $\square$

The same argument also proves $\text{Sol}(G(p, pm))$ and $\text{Sol}(G(p, pm + p - 1))$ for arbitrary integer p and natural m . Natural divisibility in Theorem 1.1 is a convenient formulation with a nonnegative quotient. The doubling construction can produce long certificates; no short-path or polynomial-complexity claim follows from the theorem.

5 Parameter consequences and the remaining induction

The formal package also includes explicit sign, period, and original-coordinate transports. One period $p \mapsto p + (2s + 1)$ uses four supplied factors of the square relator. Together with sign transport, the elementary $p = 0$ case, and Theorem 1.1, this covers the residues

$$p \equiv 0, \pm 1, \pm 2, \pm s \pmod{2s + 1}, \quad s \in \mathbb{N}.$$

In particular all integer p are covered when $s = 3$. A supplied finite 65-numbered-move residual seed, corresponding to $G(3, 4)$, completes all integer p at $s = 4$. These parameter transports and the seed are proved in `ACSsquareP2Families`, `ACSsquareP3Seed`, and `ACSsquareP3Families`. The seed's 65 moves start at its residual root pair, not at $G(3, 4)$ itself; its classical primitive cost is 81.

For reference, write the original presentation tuple as

$$P(n, a, b, c) = (x^{-1}y^nxy^{-(n+1)}, \quad x^{-1}y^axy^bx^{-1}y^c).$$

The supplied quotient-two pinches and parameter shears transfer the divisibility result to $P(2m, m, p, c)$ and $P(2m + 1, p, m + 1, c)$, for natural m, p with $p \mid m$ or $p \mid (m + 1)$ and every integer c . They also transfer the completed $s = 3, 4$ residue sets to $P(6, 3, b, c)$, $P(7, a, 4, c)$, $P(8, 4, b, c)$, and $P(9, a, 5, c)$, with all remaining parameters integral. The supplementary Lean files contain the pinch witnesses and complete transports; these statements do not assert that the entire three-parameter family at any of these exponents is solved.

Here is the precise limitation of the proposed extension. Put

$$Q_{r,d} = y^{-(r+1)}u^{-1}y^ru^{-d}.$$

Proposition 5.1 (Residual equivalence). *For $p, r \in \mathbb{Z}$ and $m \in \mathbb{N}$,*

$$\text{Sol}(G(p, pm + r)) \iff \text{Sol}((B_p, Q_{r, 2^m})).$$

Proof. Use Lemma 3.1. Conjugating A_{pm+r} by y^{pm} gives $y^{-(r+1)}u^{-1}y^r(y^{pm}u^{-1}y^{-pm})$. The inverse-word doubling witness gives $Q_{r, 2^m}$ while keeping B_p . All these ordinary paths can be reversed. $\square$

For the next residue $p = 3, r = 1$, a further supplied two-factor substitution and invertible coordinate transport give the equivalent pair (B_2, C_{2^m}) , where

$$C_d = y^{-1}u^{-1}y^{-1}u^{-1}yu^{-d}, \quad D_d = y^{-3}u^{-1}y^{-1}u^{-1}y^3u^{-d}.$$

Witnessed halving yields $(B_2, C_{2^d}) \sim_{\text{AC}} (B_2, D_d)$, but the outer y -exponents have changed. Closing the induction requires

$$(B_2, D_d) \sim_{\text{AC}} (B_2, C_d), \quad d = 2, 4, 8, \dots \quad (4)$$

The package proves that (4) for all these d is equivalent to $\text{Sol}(G(3, 3m + 1))$ for every $m \in \mathbb{N}$, using the supplied $m = 0, 1$ seeds. Neither side of this equivalence is proved here. In particular, the first missing restoration is $d = 2$; halving alone does not establish it. This is a target for further work, and even its resolution would still leave other square-family parameters to treat.

6 Formal proof and executable evidence

The package uses Lean 4.29.1 and the pinned Mathlib revision in its Lake manifest. Its copy of the official `AC.lean` is byte-identical to the SAIR source [6] at commit

99a65377c5c4f412cd9af7b8d31c41464a855736.

The main statement is `AC.MillerSchupp.SquareFamily.P2.divisibility_square_solvable`. The root witness identity and both arithmetic finishes are quantified Lean declarations, not conclusions inferred from the finite replay grid. Logical-dependency reports admit only the usual Lean axioms `propext`, `Classical.choice`, and `Quot.sound`. The conjectures are definitions of propositions and are never assumed.

The Python exporter specializes the fixed affine route, constructs the finite witnesses, and expands them into the official ordinary rank-two move table. Its root-constructor bodies were extracted unchanged from the recorded research source. An independent word interpreter maintains intermediate states, and the unchanged SAIR verifier checks the final numbered path. The exporter also handles the admissible zero case explicitly. The finite CLI has resource bounds; those are not restrictions on the quantified Lean theorem.

The shipped experiment uses 50 distinct pairs: the 44 distinct pairs (p, pm) and $(p, pm + p - 1)$ for $1 \leq p \leq 6, 0 \leq m \leq 3$; all $(2, s)$ for $0 \leq s \leq 12$; and $(0, 0)$. All 50 pass unlimited official replay. Of these, 48 satisfy the golden budgets of 100,000 numbered moves, 10,000 peak total relator length, and 5,000,000 work units. The exceptions are $(2, 11)$ and $(2, 12)$, which exceed the work budget. The recorded construction time was 2.980 seconds and replay time 0.567 seconds on the machine identified in `receipts/replay.json`; these are one-run measurements, not portable performance guarantees.

p	s	Numbered moves	Official work	Peak length
2	0	183	3,185	35
2	3	282	9,215	83
2	6	637	58,022	273
2	10	5,457	2,766,856	1,273
2	11	19,176	14,902,075	1,779
2	12	21,819	26,190,511	2,829

The six exact overlaps with frozen search certificates all replay independently. The uniform constructor improves none of their lengths. Individual solved instances in the earlier comparison were already represented in the public Caltech solved materials; the computational evidence here is reproduction and validation, not a claim of newly solved benchmark challenges [5]. Certificate counts, witness-factor counts and primitive costs measure different objects and are not interchanged.

From the repository root, the reproduction commands are

```
python3 verify.py --lean
python3 experiment.py --matrix --output run/replay.json
```

Lean and the pinned Mathlib dependencies must first be available as described in the README. The package needs no model weights, search run, sibling research directory, or credentials. The path exporter and the Lean proof are separate implementations; identical complete witness lists between them are not asserted. Nine regression tests cover all numbered move types, donor restoration, the affine entry at signed parameters, witness corruption, boundary inputs, and equality with all 49 previously recorded constructed paths.

Provenance and assistance. The research source is frozen at commit

68ceae3d422eee6de28bfd54793b21b2031cbda3.

The accompanying `SOURCE_PROVENANCE.json` records the copied and extracted source hashes. This work was developed with Codex and Fable assistance for symbolic exploration, coding, proof development, and review. Fable’s preserved square-family analysis identified the quotient-two pinch and the earlier $p = \pm 1$ connection to the published $MS(1, w)$ theorem. The finite baseline certificates are credited to the earlier solved materials, not to the uniform construction. Human author and release review remain appropriate before public circulation of this draft.

A A complete affine entry route

In the following table, slots are indexed by 0, 1. I_i inverts relator i , M_i^ϵ replaces it by $R_i R_{1-i}^\epsilon$, and $C_i(g)$ conjugates it by g . An M_i^{-1} entry expands into inversion of the donor, multiplication, and inversion of the donor again. Apply the rows in order to $G(p, s)$. Rows 1–12 reach the normal form in Lemma 3.1; rows 13–51 reach $f(B_p, A_s)$. Thus the 51 macros comprise 61 primitives. The table is generated from the same frozen affine route distributed as `entry_route.json`; it holds for all integer p, s .

Row	Operation
1	$C_0(u^{-2})$
2	$C_1(u^{-1}y^{-p}u^{-1})$
3	M_0^{-1}
4	$C_0(y^s)$
5	M_1^{-1}
6	$C_1(y^s)$
7	I_1
8	$C_0(y)$
9	$C_1(y)$
10	M_0^{-1}
11	I_0
12	$C_0(u^{-1})$
13	I_0
14	$C_0(y^p u^{-1} y^s u)$
15	I_1
16	$C_1(y^p u y^{-1})$
17	M_0^1
18	I_0
19	M_1^1
20	I_0
21	$C_0(y^p u^{-1} y^s u y^{-p} u^{-1})$
22	M_0^1
23	I_1
24	$C_1(y^p u y^{-1} u y^{-p} u^{-1})$
25	M_0^{-1}
26	$C_1(u y^p u^{-1} y u^{-1} y^{-p})$
27	$C_0(y^{-p})$
28	$C_1(y^{-p+s} u y^{-p} u^{-1})$
29	M_0^1
30	I_0
31	M_1^1
32	I_0
33	$C_0(u y^{-s-1} u y^{-p} u^{-1} y^p)$
34	M_0^1
35	I_1
36	$C_1(y^{-p+s} u y^{-p} u^{-1} y^p)$
37	M_0^{-1}
38	$C_1(y^{-p} u y^p u^{-1} y^{p-s})$
39	I_0
40	$C_0(y^{-s})$
41	$C_1(y^{-2p+s+1} u^{-1} y^p)$
42	M_0^1
43	I_0
44	M_1^1
45	I_0
46	$C_0(y^{-p-s} u^2 y^{p-s-1})$
47	M_0^1
48	I_1
49	$C_1(y^{-p+s+1} u^{-1} y^{p+s})$
50	I_1
51	$C_0(y^p)$

References

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